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Paper Code : CS703C ARTIFICIAL INTELLIGENCE

Time Allotted : 3 Hours

Full Marks :70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[1 x 10 = 10]

- (I) What is information retrieval?
- (II) How to test whether a machine can act humanly or not?
- (III) Are reflex actions (such as flinching from a hot stove) rational? Are they intelligent?
- (IV) What is a tautology?
- (V) List a few applications of Bayesian networks.
- (VI) What is rote learning?
- (VII) Design the PEAS information for Automated Car Agent.
- (VIII) Give examples of competitive and cooperating agents?
- (IX) Differentiate declarative and procedural knowledge.
- (X) What is hierarchical planning?
- (XI) Give an example of an environment which is competitive and cooperating at the same time?
- (XII) Differentiate between Uninformed and Informed Search techniques.

Group-B (Short Answer Type Question)

Answer any three of the following

[5 x 3 = 15]

2. Discuss Bidirectional Search technique. [5]
3. What are fuzzy and crisp sets ? State their differences. [5]
4. Using constraint satisfaction problem, solve the following cryptarithmic problem: [5]
 CROSS
 ROADS

 DANGER
5. Derive the parse tree for the sentence "John ate the cake" [5]
 $S \rightarrow NP VP$
 $NP \rightarrow N \mid DET N$
 $VP \rightarrow VG NP$
 $DET \rightarrow a \mid an \mid the$
 $VG \rightarrow V$
 $V \rightarrow ate$
 $N \rightarrow John \mid cake$
6. State different problems in hill climbing algorithm. [5]

Group-C (Long Answer Type Question)

Answer any three of the following

[15 x 3 = 45]

7. Write short notes on the following topics: (a) tautology, (b) contradiction, and (c) proposition. [15]
8. Write short notes on the following topics: (a) alpha-beta cutoff, (b) zero-sum game, and (c) adversarial search. [15]
9. Write short notes on the following topics: (a) Utility-based agents, (b) Goal-based agents, and (c) Properties of task environments. [15]
10. Illustrate how Depth Limited Search and Iteratively Deepening Depth First Search improves the performance of Depth First Search algorithm. [15]

11. The following table contains information on the employees from a small private firm. Considering 'Appraisal indication' as the class label attribute, construct a decision tree from the data. [15]

Department	Employee ID	Employee Name	Salary	Experience	Appraisal indication
Sales	001	Roy	350	4	No
Sales	002	Sen	400	3	Yes
Technical	004	Mahato	500	2	No
Technical	008	Sharma	600	2	Yes
Technical	010	Dubey	600	2	Yes
HR	011	Cruz	400	3	No
HR	017	Iyer	300	1	No

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